International and Monetary Economics

Course Code: 30054

Professor: Dmitriy Sergeyev

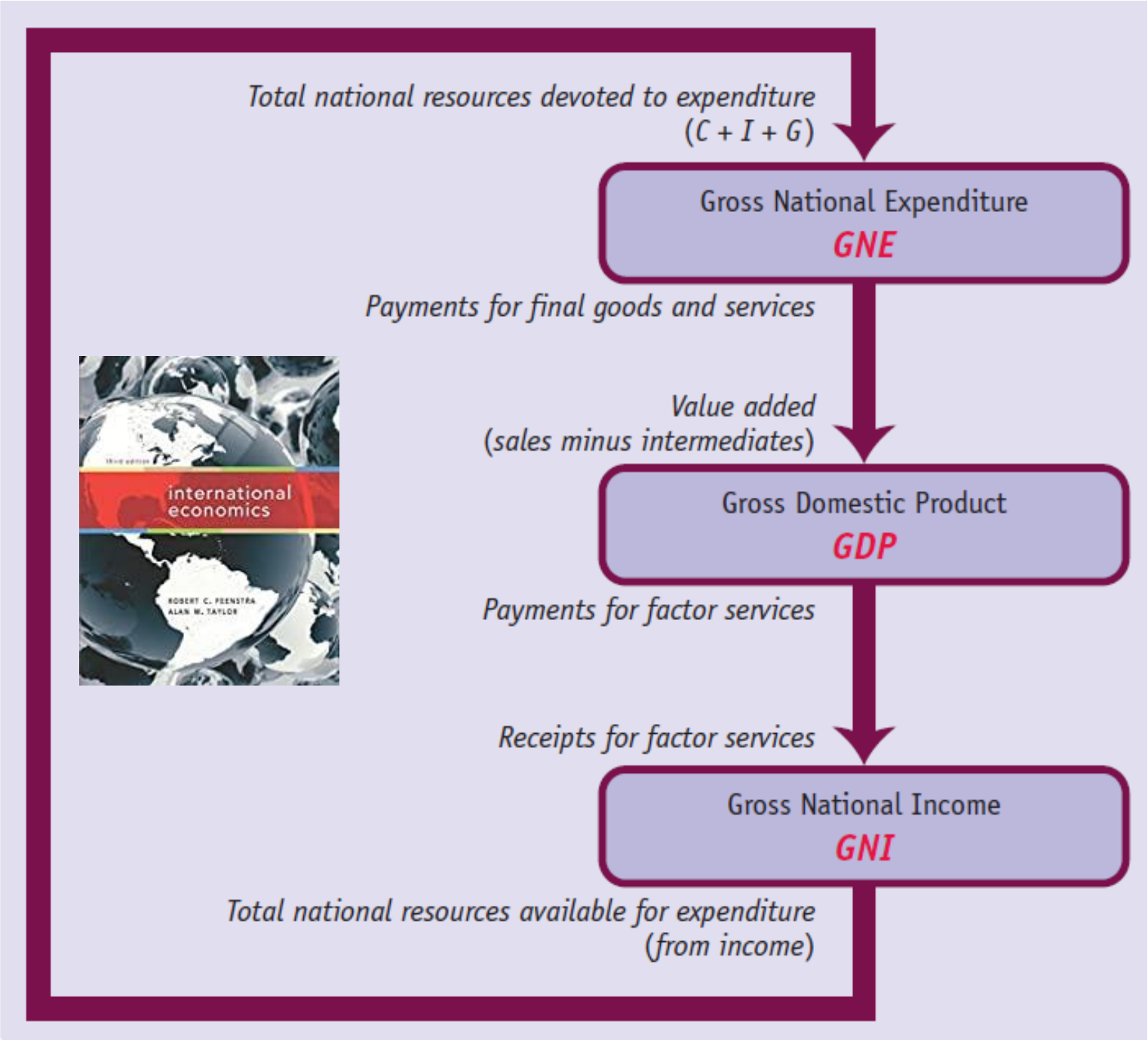
Lecture 2

Roadmap

- International income accounting
  1. Lots of definitions and examples
  2. Current account and its interpretations
     - Saving and investment
     - Net international investment position

Aggregate Economic Activity

An Overview



National Income and Product Accounts (**closed economy**)

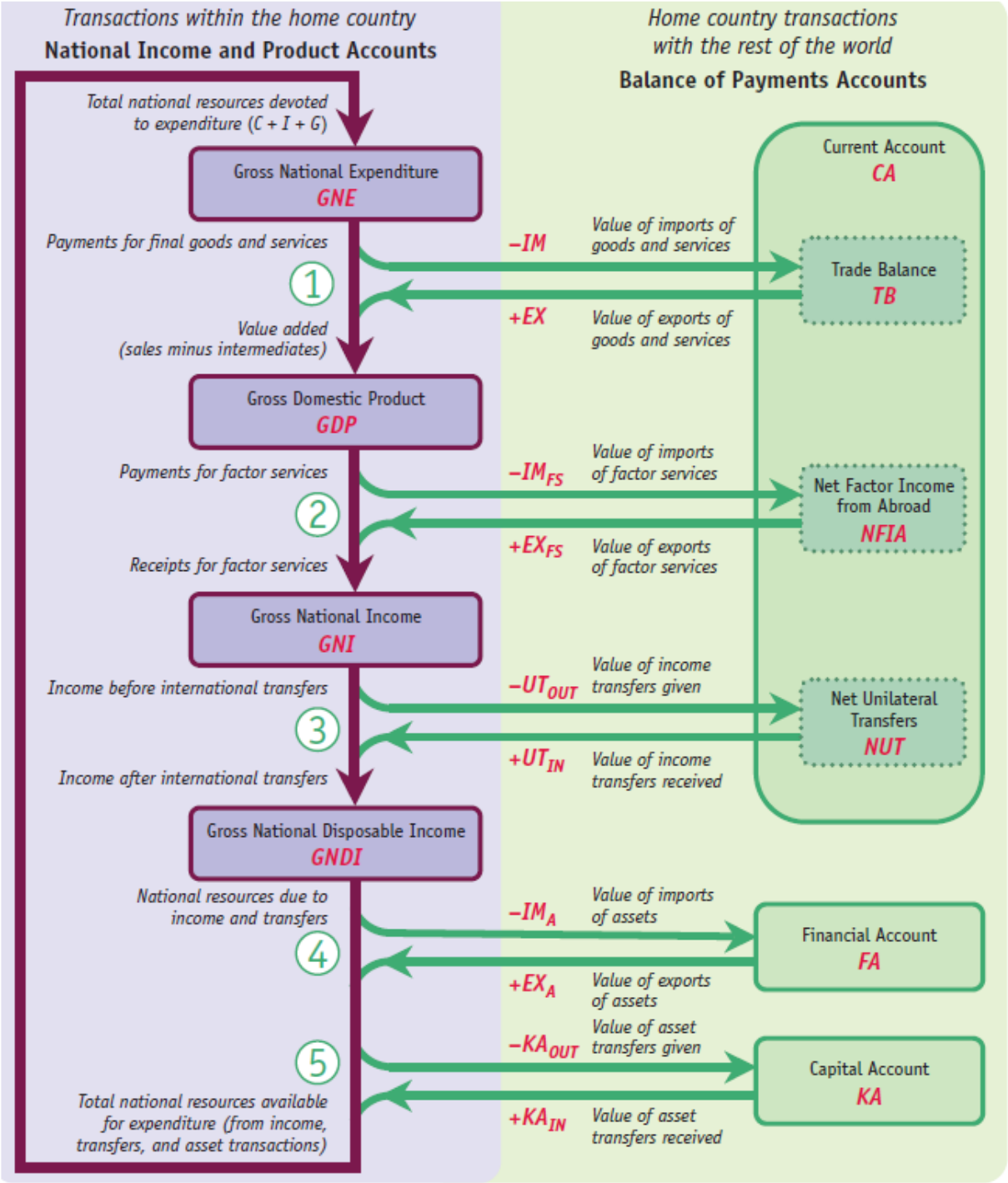
- GNE – expenditure approach
- GDP – product approach
- GNI – income approach

In a closed economy

- **GNE=GDP=GNI**

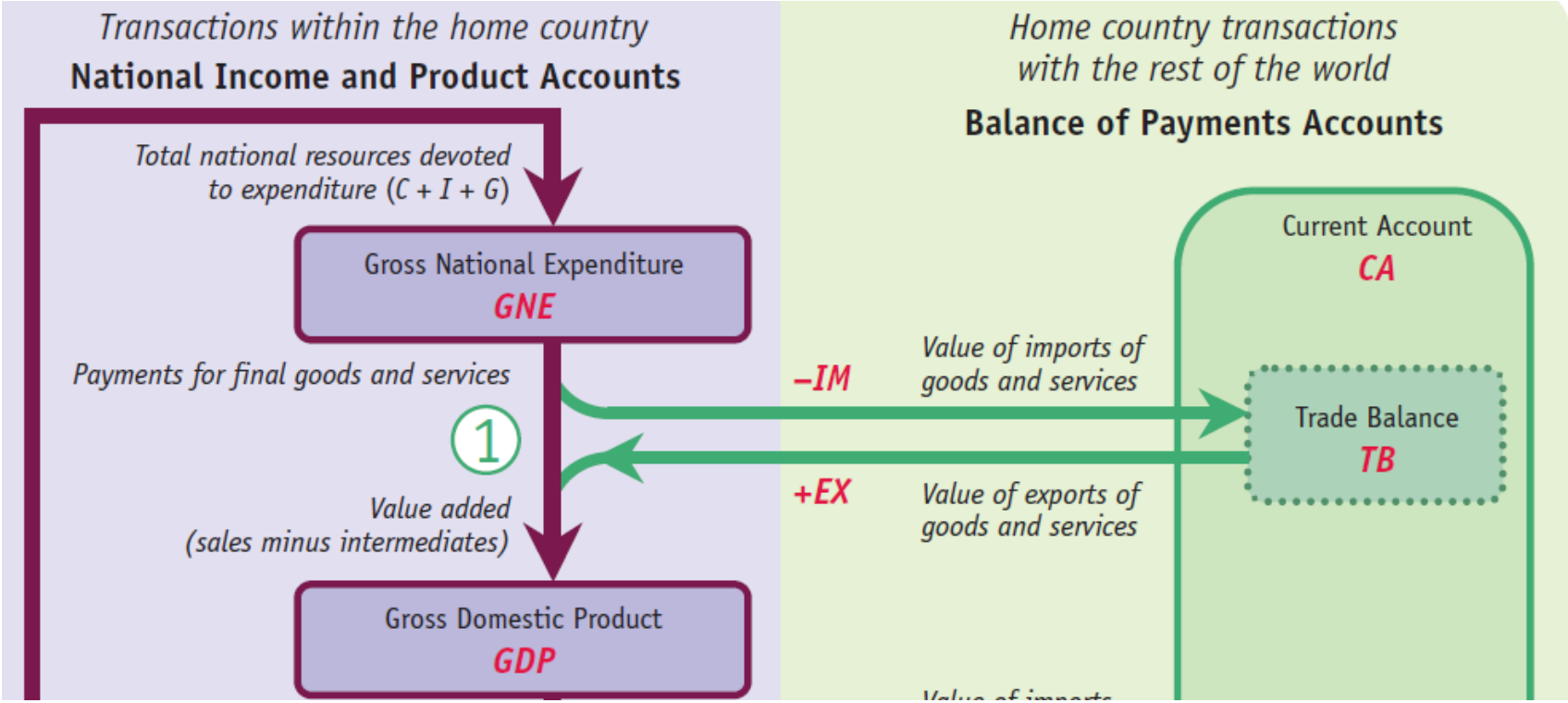
Source: International Economics, Chapter 16, Feenstra and Taylor, 3rd edition

The Flow of Payments in an **Open Economy**



Balance of payments (BOP) accounts adds all of the extra payment flows to and from the rest of the world

Step 1: from GNE to GDP



GNE = C+I+G

GDP=(C+I+G)+Ex-Im

Ex – exports, Im – imports

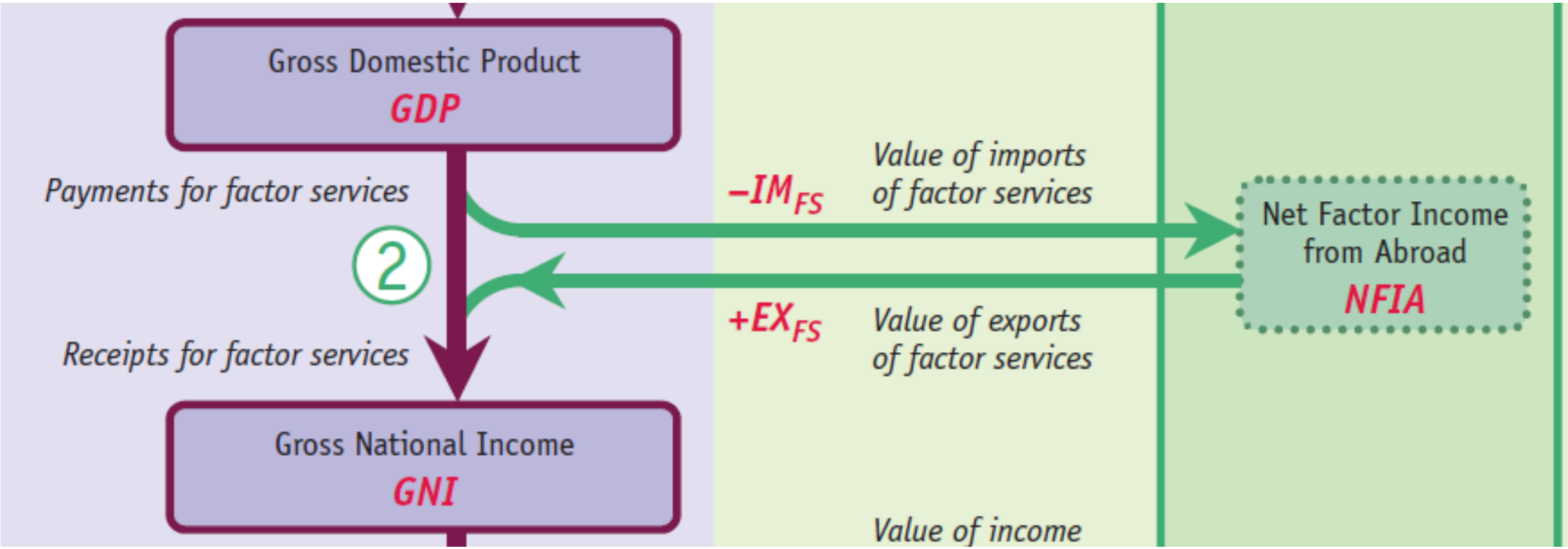
Ex - Im – **trade balance** or **net exports**

Step 2: from GDP to GNI

GNI = GDP + NIFA

Note:

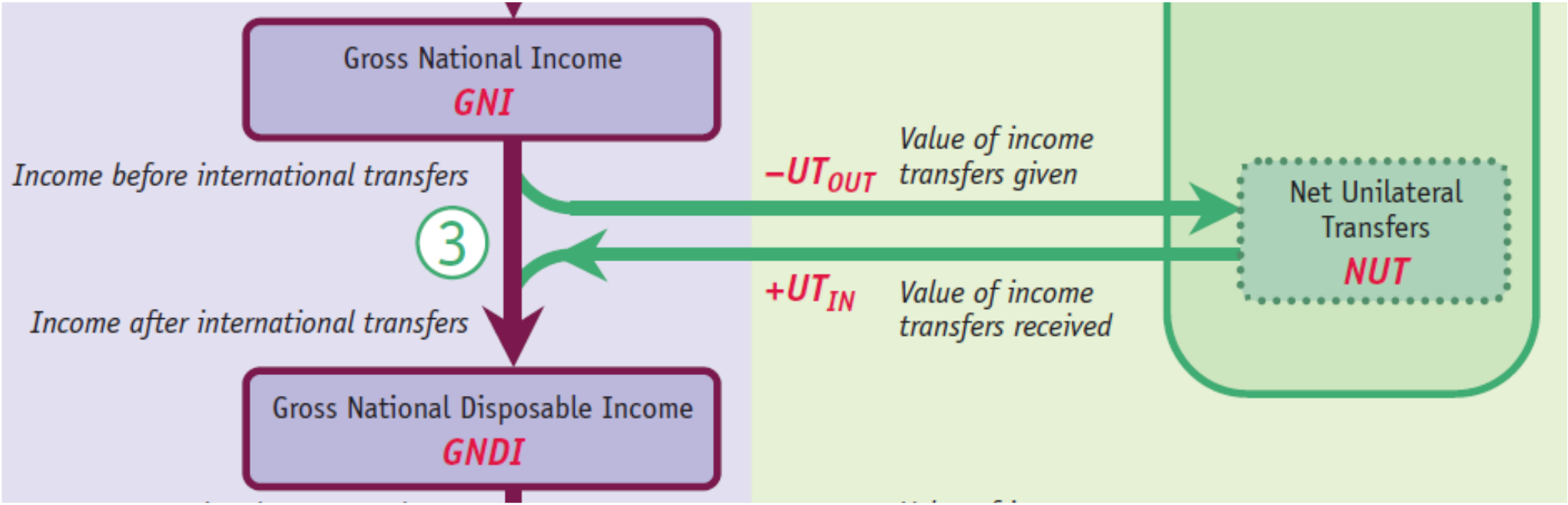
- NIFA (**N**et factor **I**ncome **F**rom **A**broad) = NFIA (**N**et **F**actor **I**ncome from **A**broad)
- GNI and GNP (gross national product) are the same things (according to the World Bank definition)



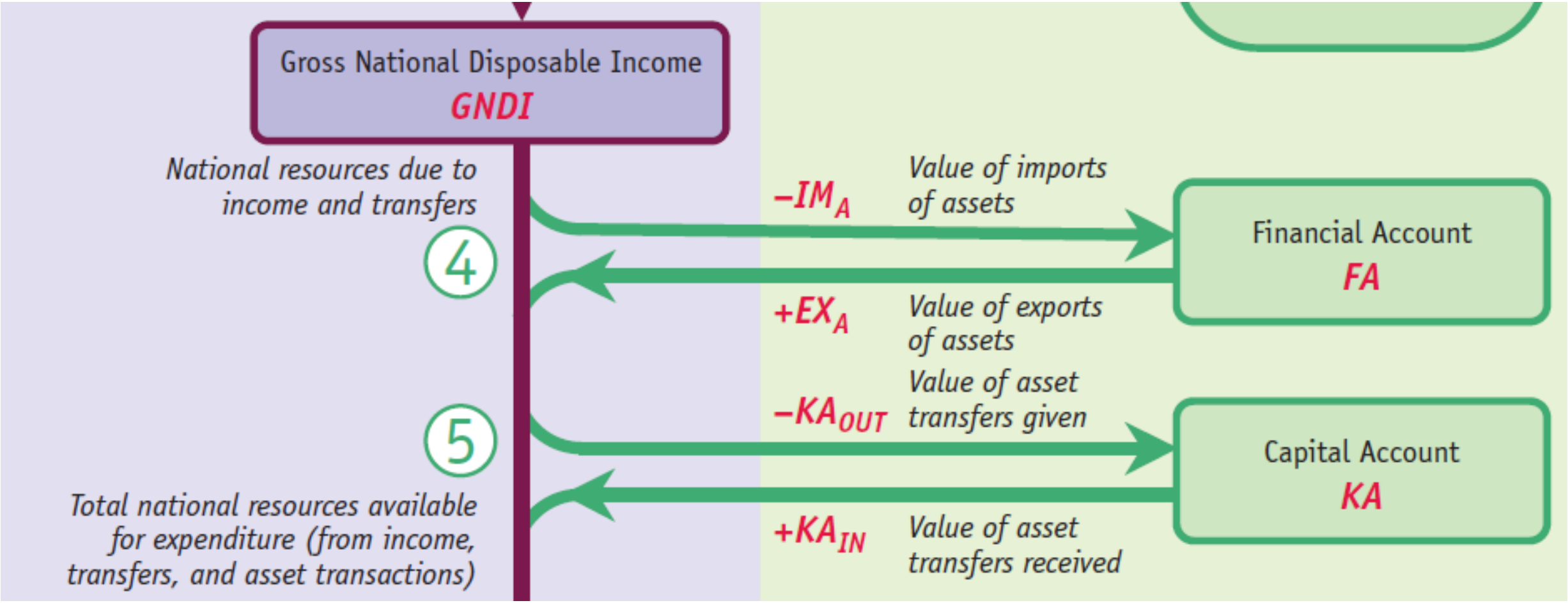
Step 3: from GNI to GNDI

- Unilateral transfers – non-market transactions
  - foreign aid
  - remittances by migrants to their families back home
  - gifts from abroad

GNDI = GNI + NUT

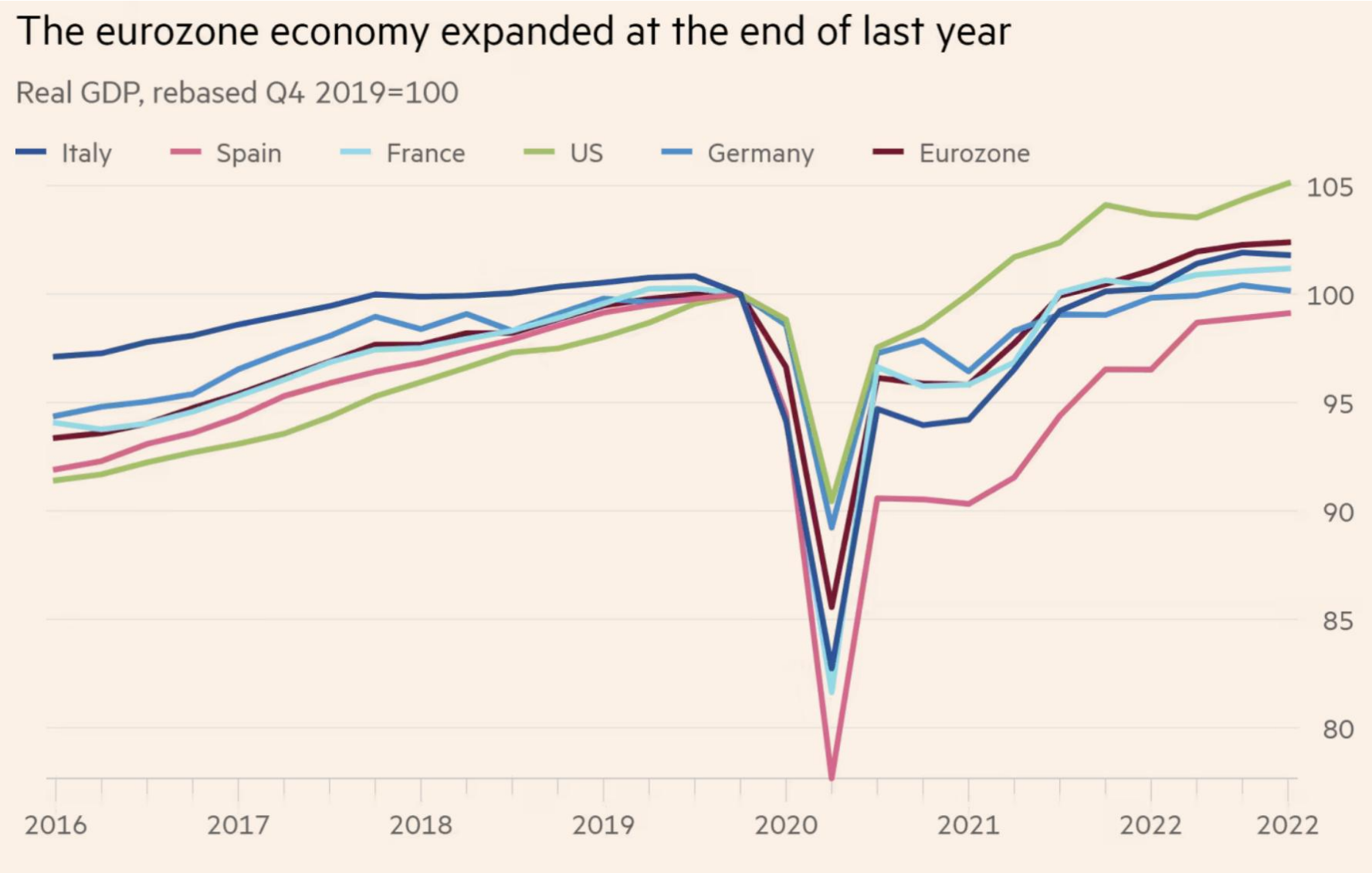


Steps 4 and 5: from GNDI to GNE



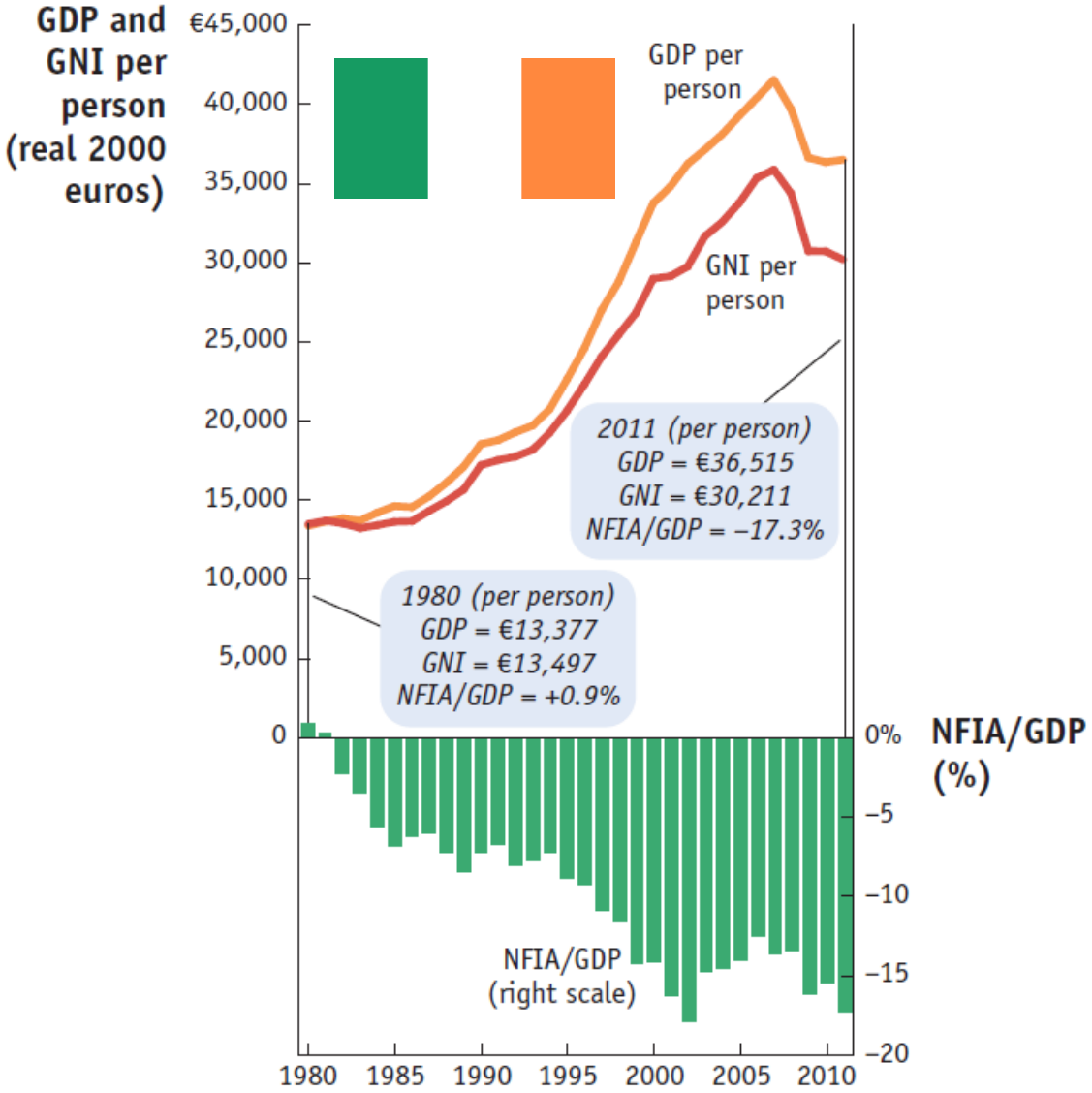
Examples

Example 1



Source: FT

Example 1: GNI vs. GDP: The Celtic Tiger



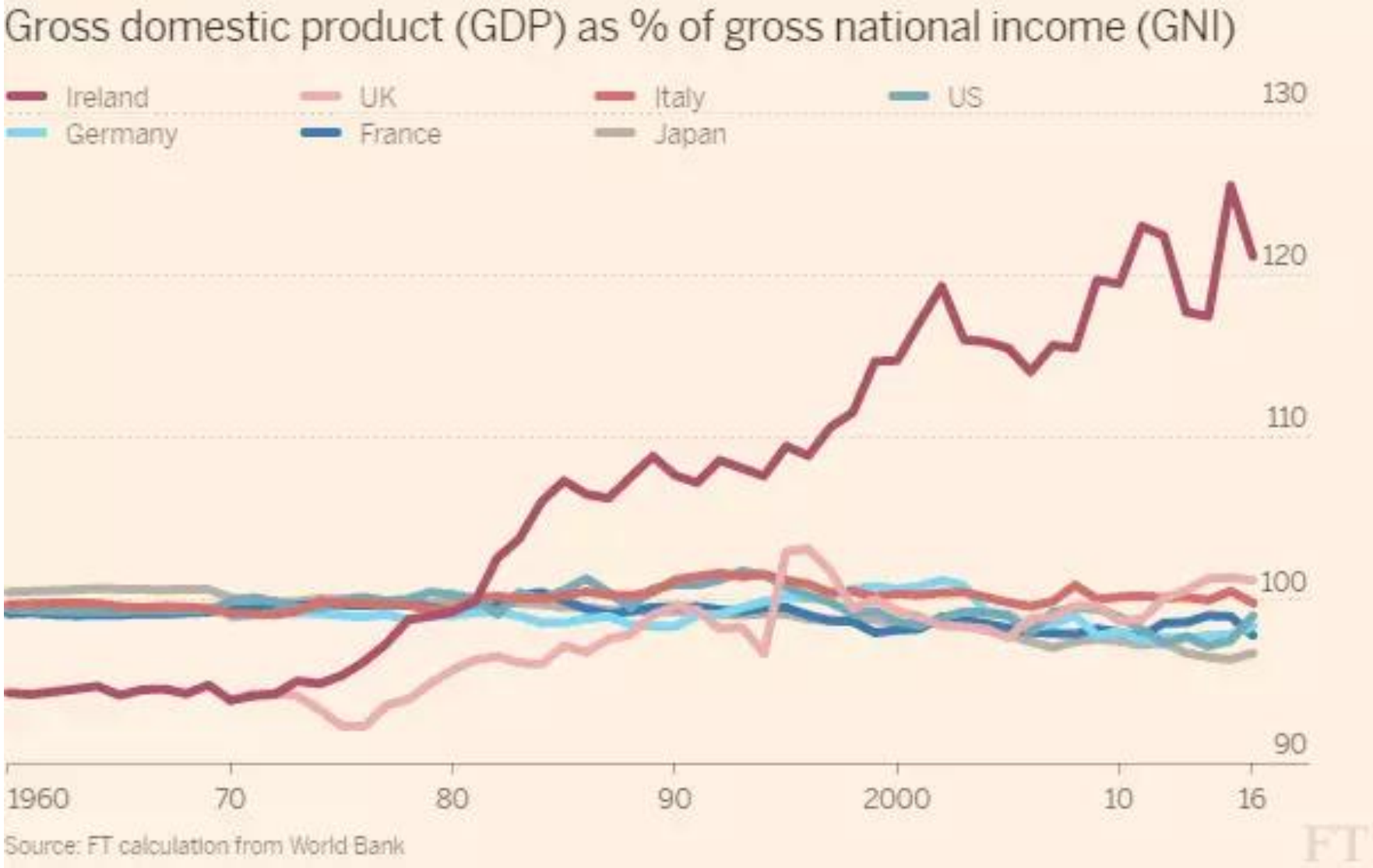
Background

- Ireland was one of the poorest countries in Europe in the 1970s
- The Irish Miracle: 1970-2008
  - Profit shifting may account for part of high GDP growth

Did Irish citizens enjoy *all* the gains?

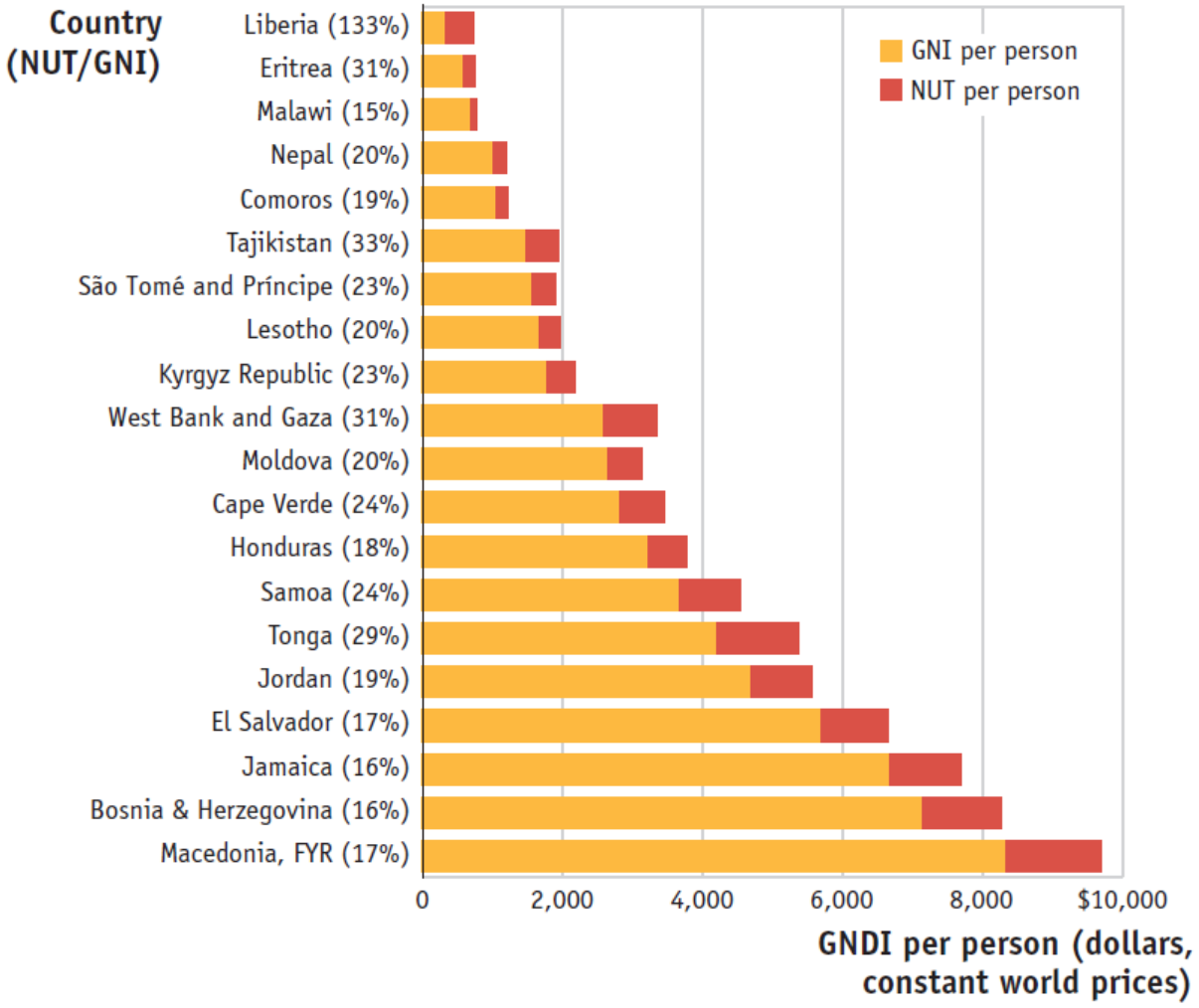
- 75% of industrial-sector GDP was in foreign-owned plants in 2004
- NIFA/GDP>15% after 2000
- Ireland was 4th richest OECD country by GDP pc in 2004, but 17th by GNI pc

Example 1: GNI vs. GDP: Other Countries



- Differences between GDP and GNI are usually negligible.
- Some exceptions
  - Ireland (again): GDP/GNP = 1.2 (2016) due to foreign firms
  - Luxembourg: GDP/GNP = 1.5 (2017) due to foreign workers

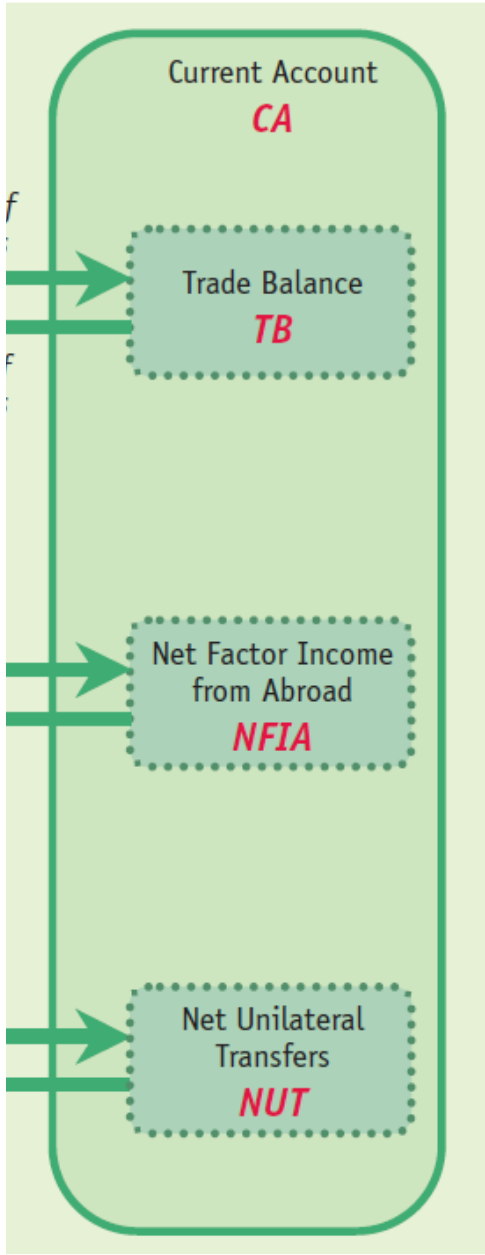
Example 2: GNDI =GNI+NUT

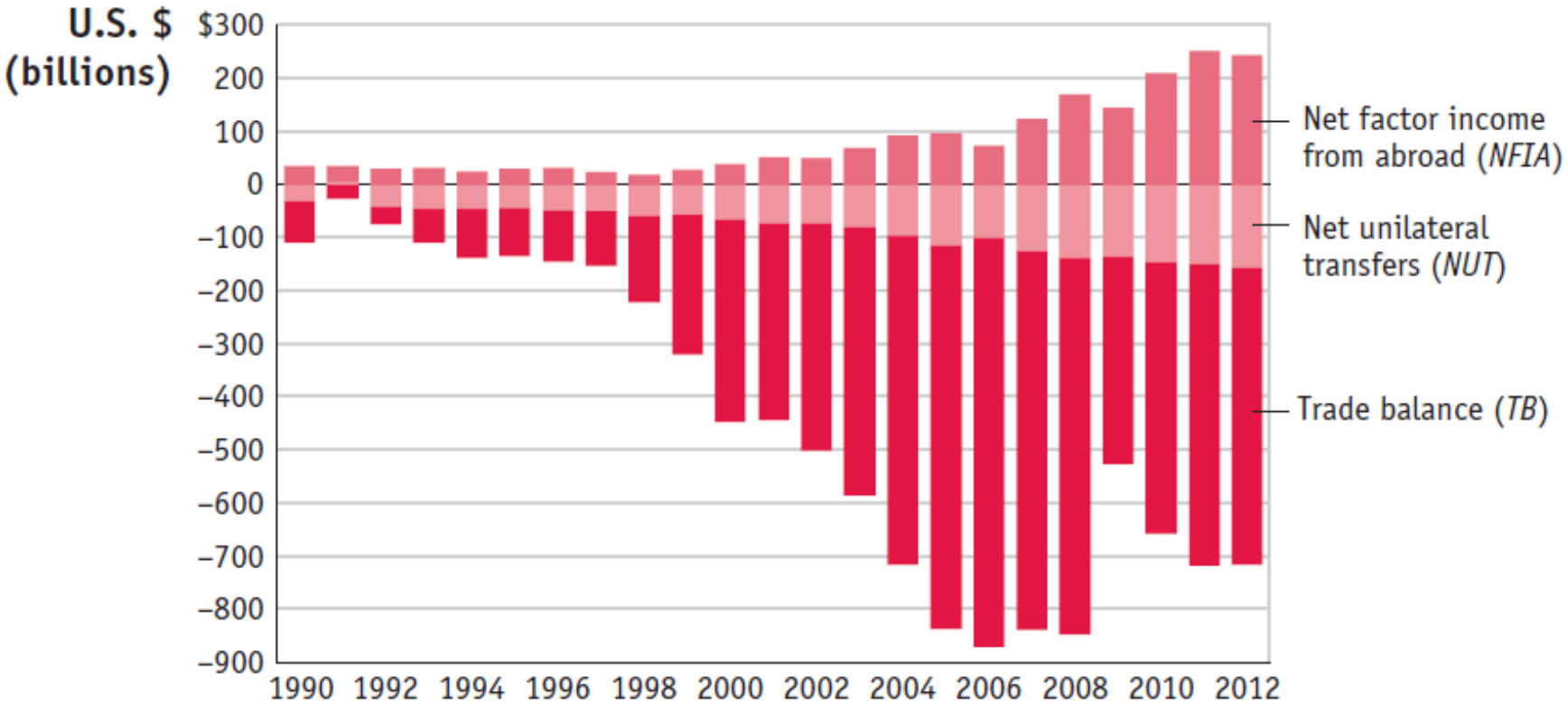


- Countries with NUT/GNI>0.15 during 2001-10
- Importantly, NUT include both government and private transfers
  - US private NUT was three times more than government NUT in 2000

Example 3: CA=NX+NIFA+NUT

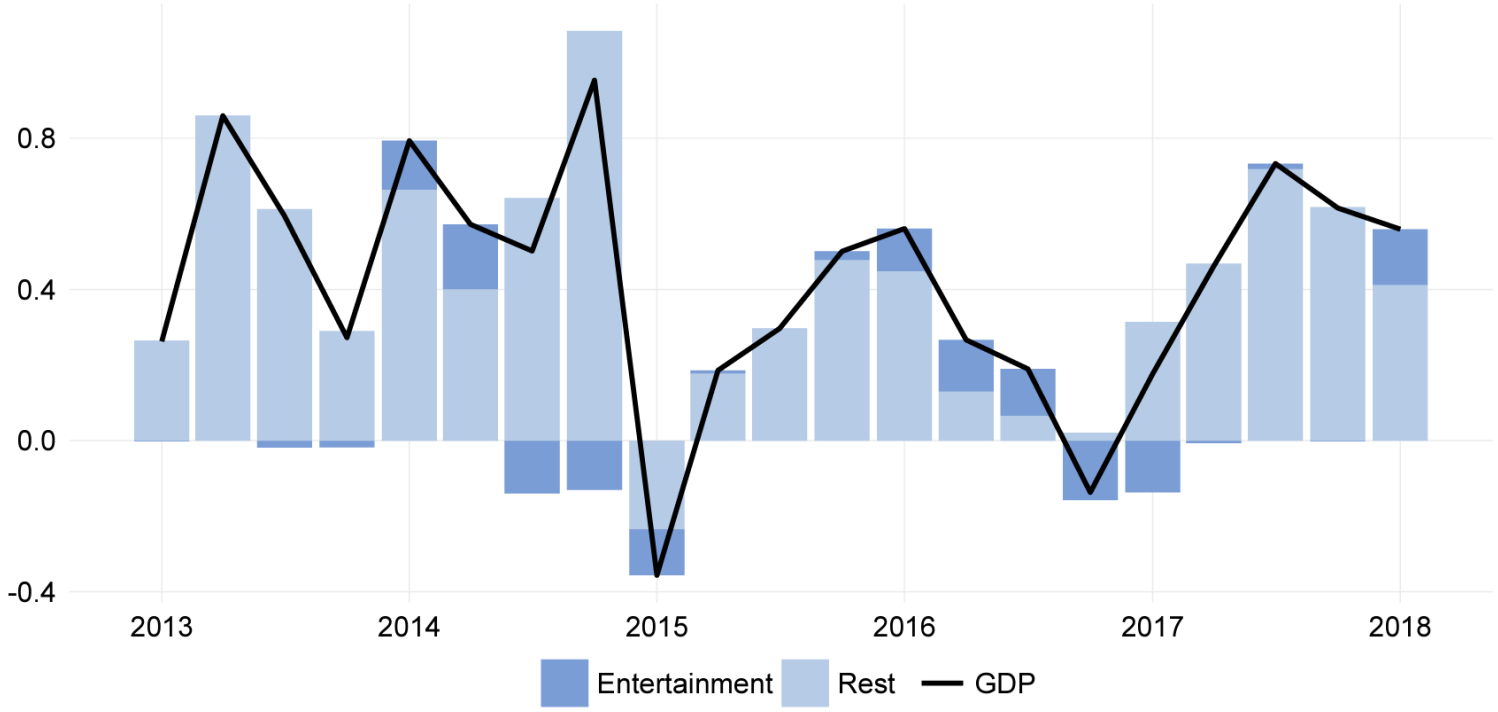
in the US





Example 4: How Big Sport distorts Swiss GDP

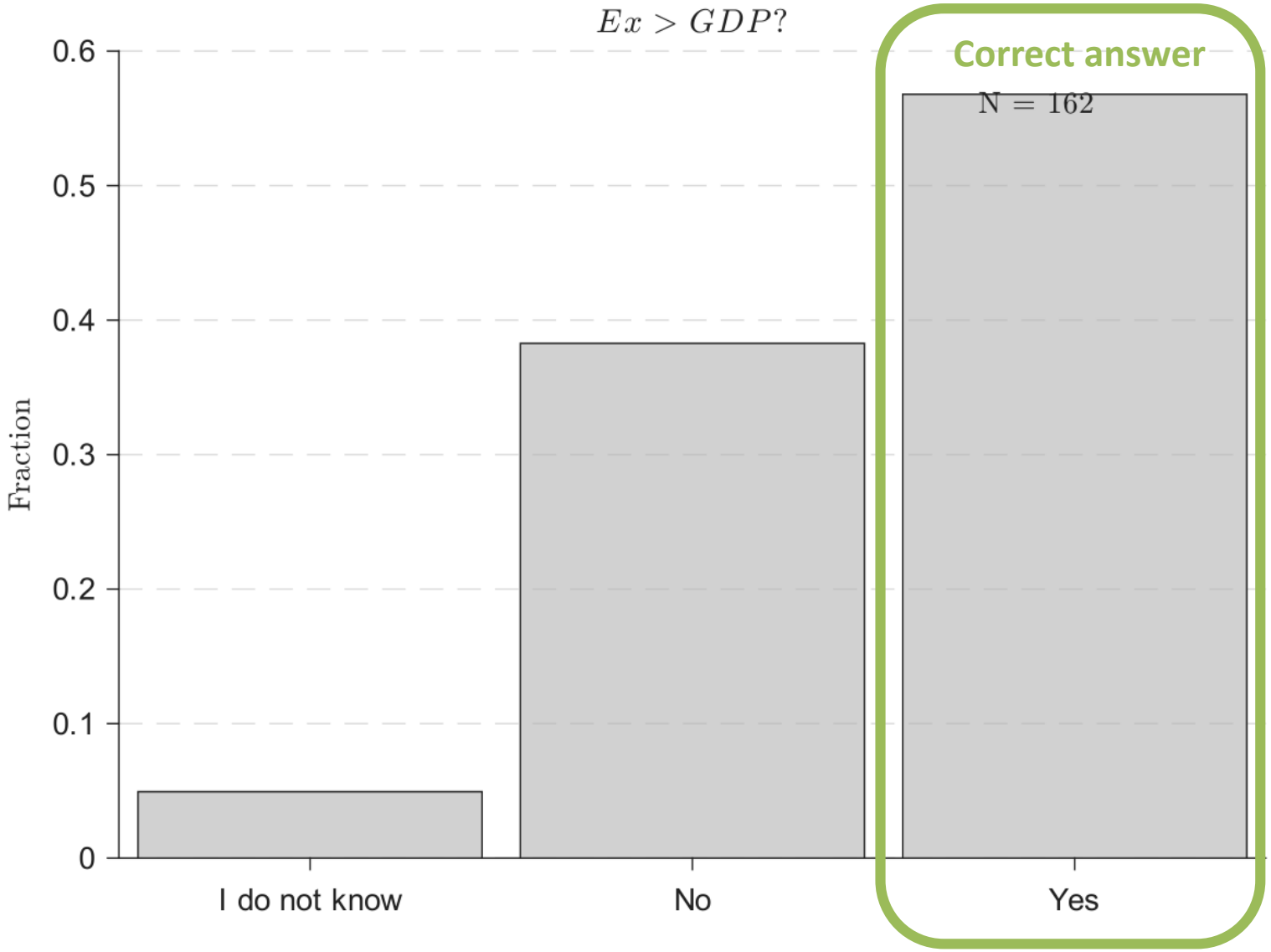
Figure 1: Gross domestic product
Real, seasonally and calendar adjusted, change to previous quarter in %, growth contributions in percentage points



- FIFA, UEFA, and IOC are located in Switzerland.
- Their revenues are connected to the organization and marketing of major sporting events.
- This is income from intangible assets (licenses, rights, patents, etc.).

Source: SECO

Example 5: Exports and GDP

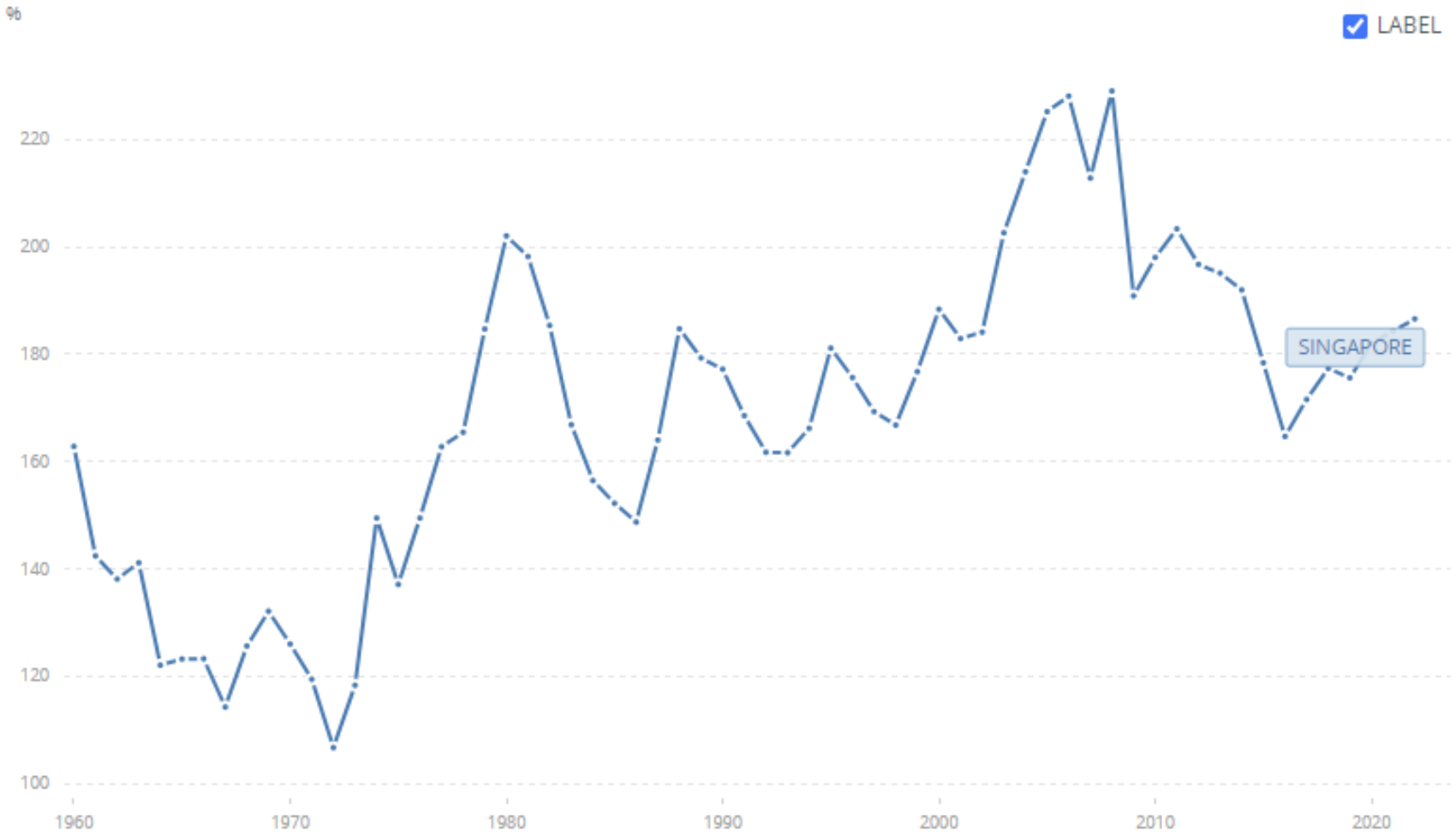


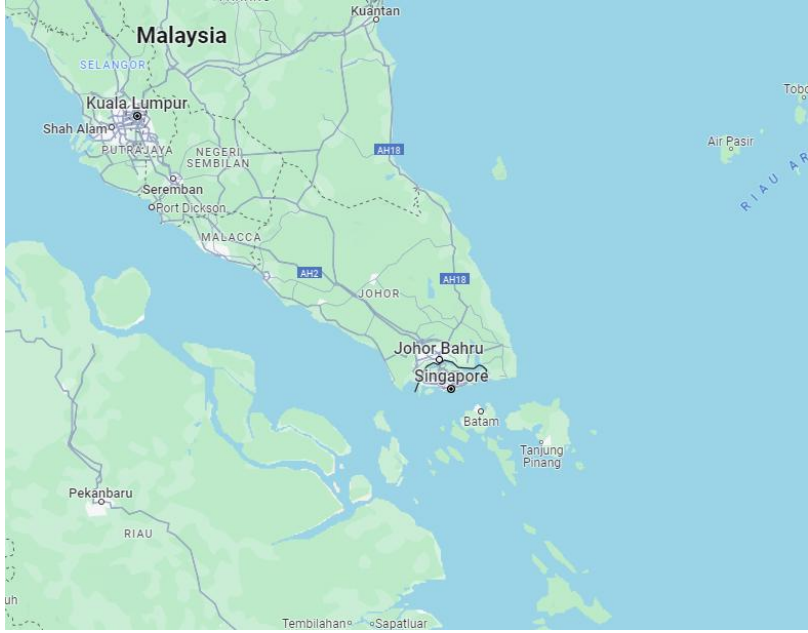
Exports of goods and services (% of GDP) - Singapore

World Bank national accounts data, and OECD National Accounts data files.

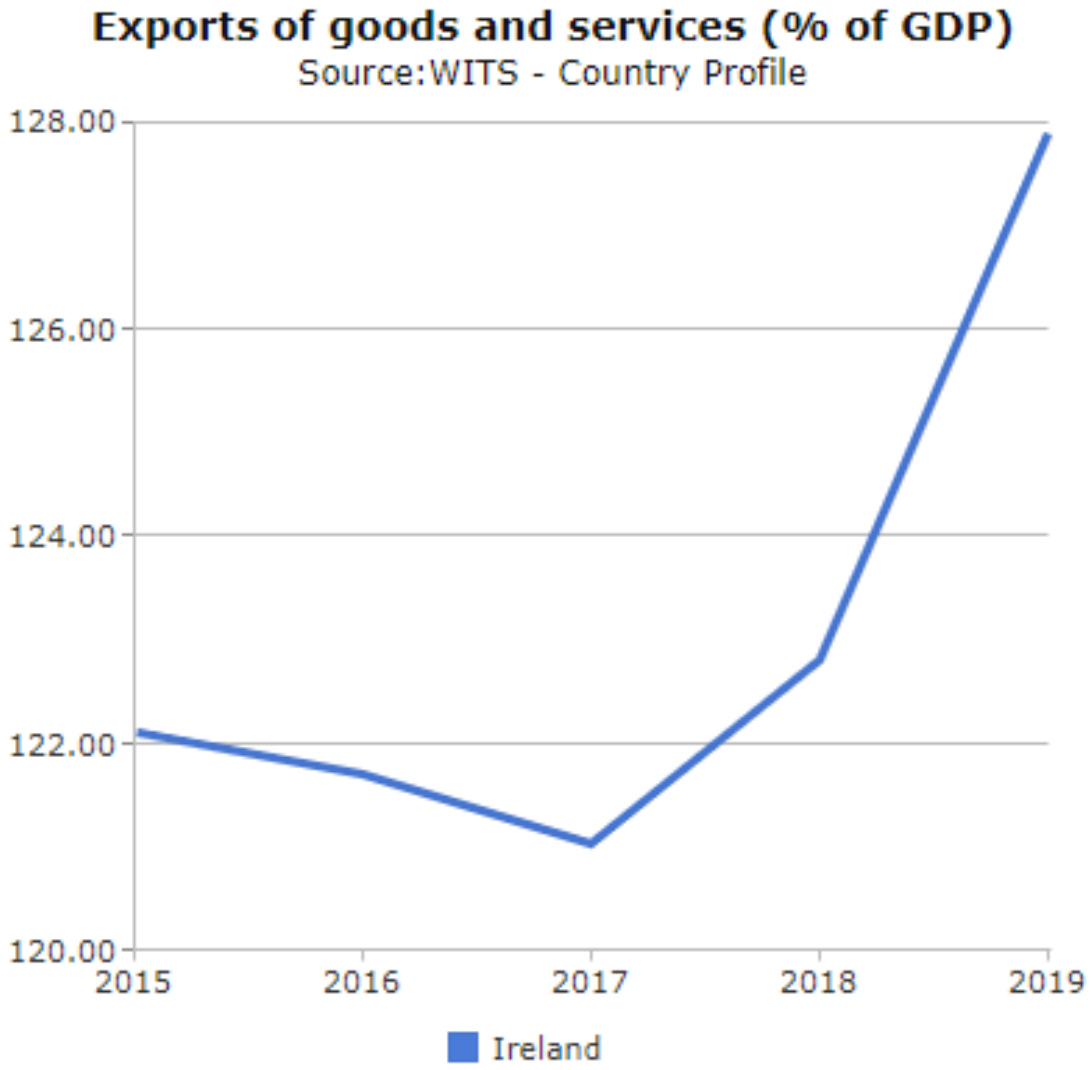
License : CC BY-4.0

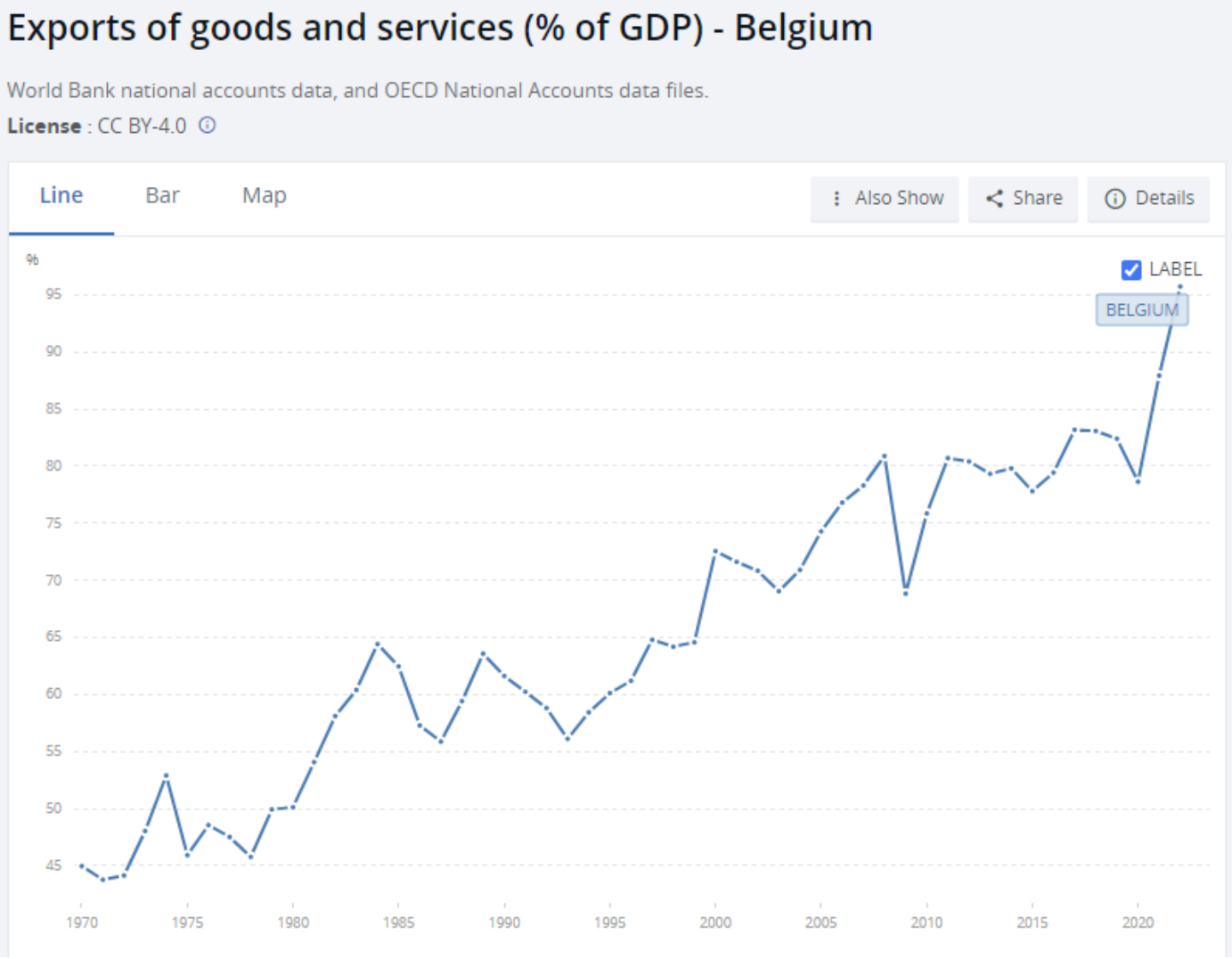
Line Bar Map Also Show Share Details





Singapore is a regional hub that imports and re-exports lots of goods





Ireland and Belgium are European regional hubs that import and re-export lots of goods.
(Note: Belgium has not crossed the 100-percent mark yet)

Example 6: Trade balance and GDP

This is a theoretical possibility that tests your understanding that investments can be negative.

- Imagine that the country does not produce anything this year (GDP = 0) and does not import from the rest of the world.
- Instead, it dismantles existing capital (negative investment) to sell it abroad (positive exports). Hence, NX>0 and GDP=0

In 2022, no country had NX>GDP

Countries	Trade balance, percent of GDP, 2022	Global rank
Gabon	44.78	1
Ireland	39.91	2
Qatar	36.9	3
Singapore	36.24	4
Luxembourg	34.04	5
Azerbaijan	33.03	6
R. of Congo	29.46	7
Norway	28.35	8
Bermuda	26.37	9
Brunei	25.8	10

Current Account

Current Account Interpretations

1. CA, Saving and Investment

2. CA and NIIP

Current Account, Saving and Investment

- An open-economy **national income identity**

$$GNDI = C + I + G + CA$$

- By definition, saving is the part of income that is not consumed

$$S = GNDI - (C + G)$$

  - **Language**: in Economics, "saving" is a flow variable and "savingS" is a stock variable

- Rearranging the national income identity, we get

$$CA = (GNDI - C - G) - I$$

or

$$CA = S - I \quad \Leftrightarrow \quad S = I + CA$$

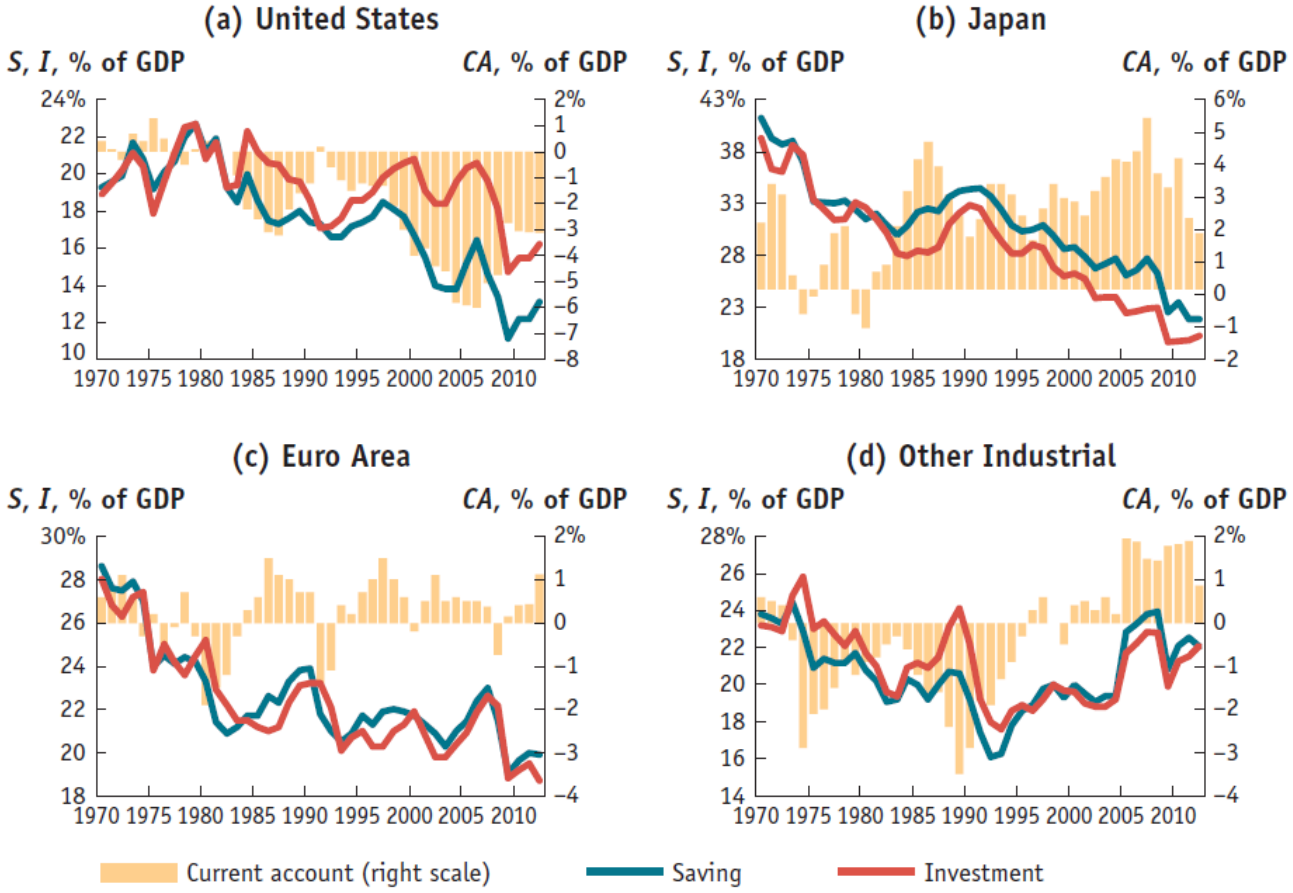
Current Account, Saving and Investment

$$S = I + CA$$

- a country can save more than it invests ($S > I$), by having a current account surplus ($CA > 0$) (i.e., by landing to the rest of the world)

- a country can invest more than it saves ($I > S$), by having a current account deficit ($CA < 0$) (i.e., by borrowing from abroad)

Current Account, Saving and Investment



- Saving/GDP fell in most industrialized countries
- Investment behavior differs across countries
- Current account automatically reflects these differences

Current Account, Saving and Investment

Decomposing Saving

- Private saving (part of **private** disposable income that is saved rather than consumed)

$$S^P = (GNDI - T) - C$$

- Government saving (extent to which the government is borrowing to finance its expenditures)

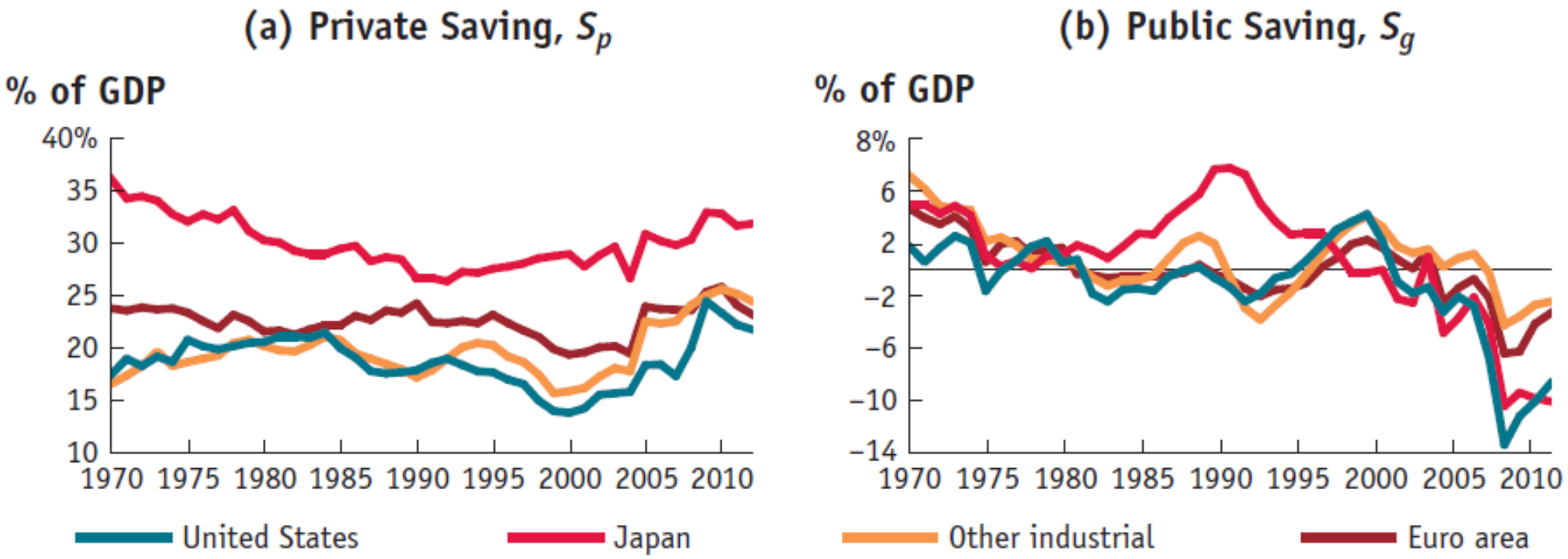
$$S^G = T - G$$

- National saving

$$S = S^P + S^G = I + CA$$

Current Account, Saving and Investment

Decomposing Saving



Current Account, Saving and Investment

Decomposing Saving

Question

- Do government deficits cause current account deficits?

CA and Net International Investment Position

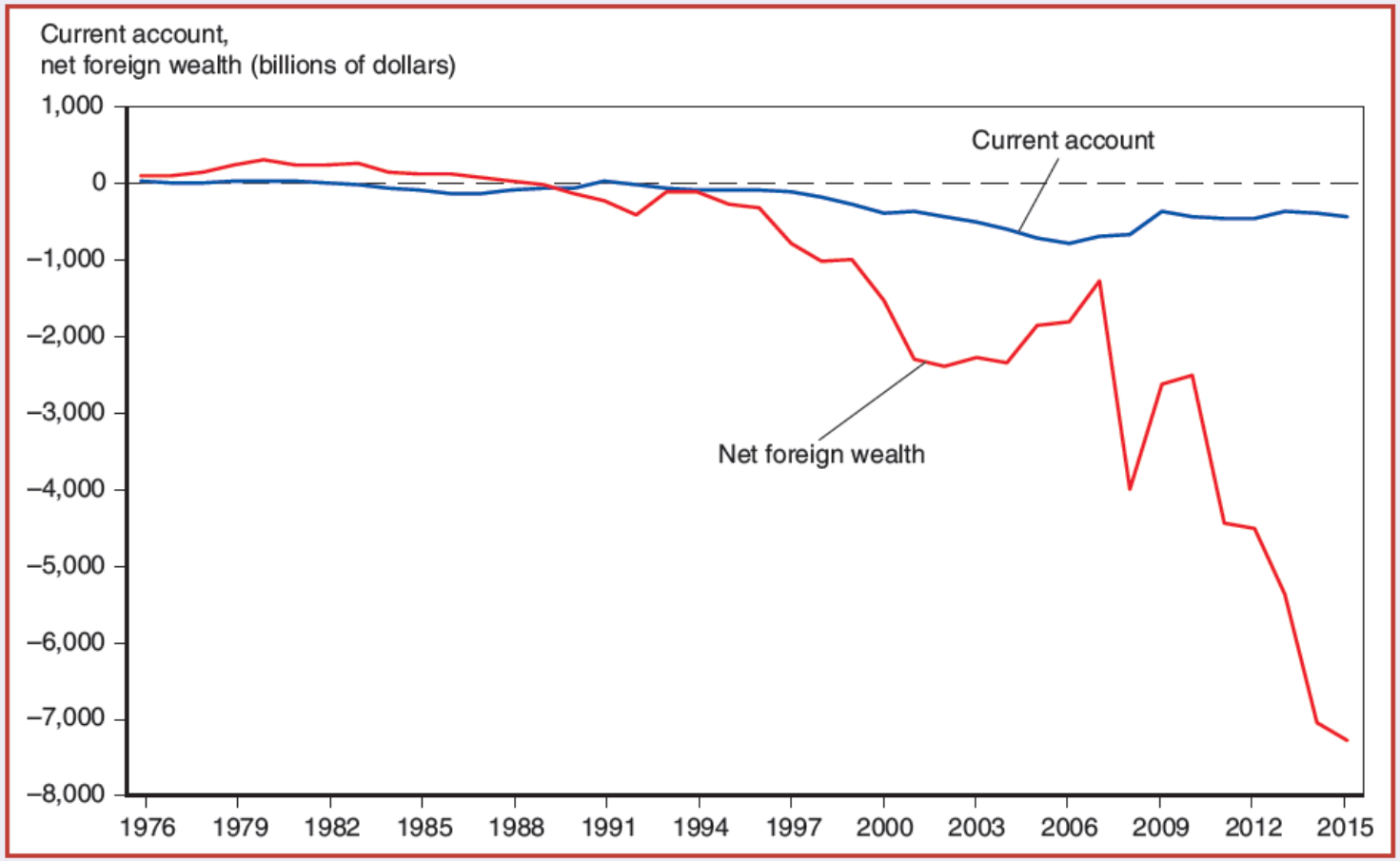
- *Net* international investment position (NIIP) is the difference between the value of foreign assets held by domestic residents and domestic assets held by foreign residents (i.e., domestic liabilities)

- Current account results in a change in NIIP

$$\Delta NIIP_t = NIIP_t - NIIP_{t-1} \approx CA_t$$

  - $NIIP_t$- the country's *net* international investment position at the end of period t
  - An approximation is due to *the valuation effect* (coming soon)

CA and Net International Investment Position



Pearson MyLab Economics Real-time data

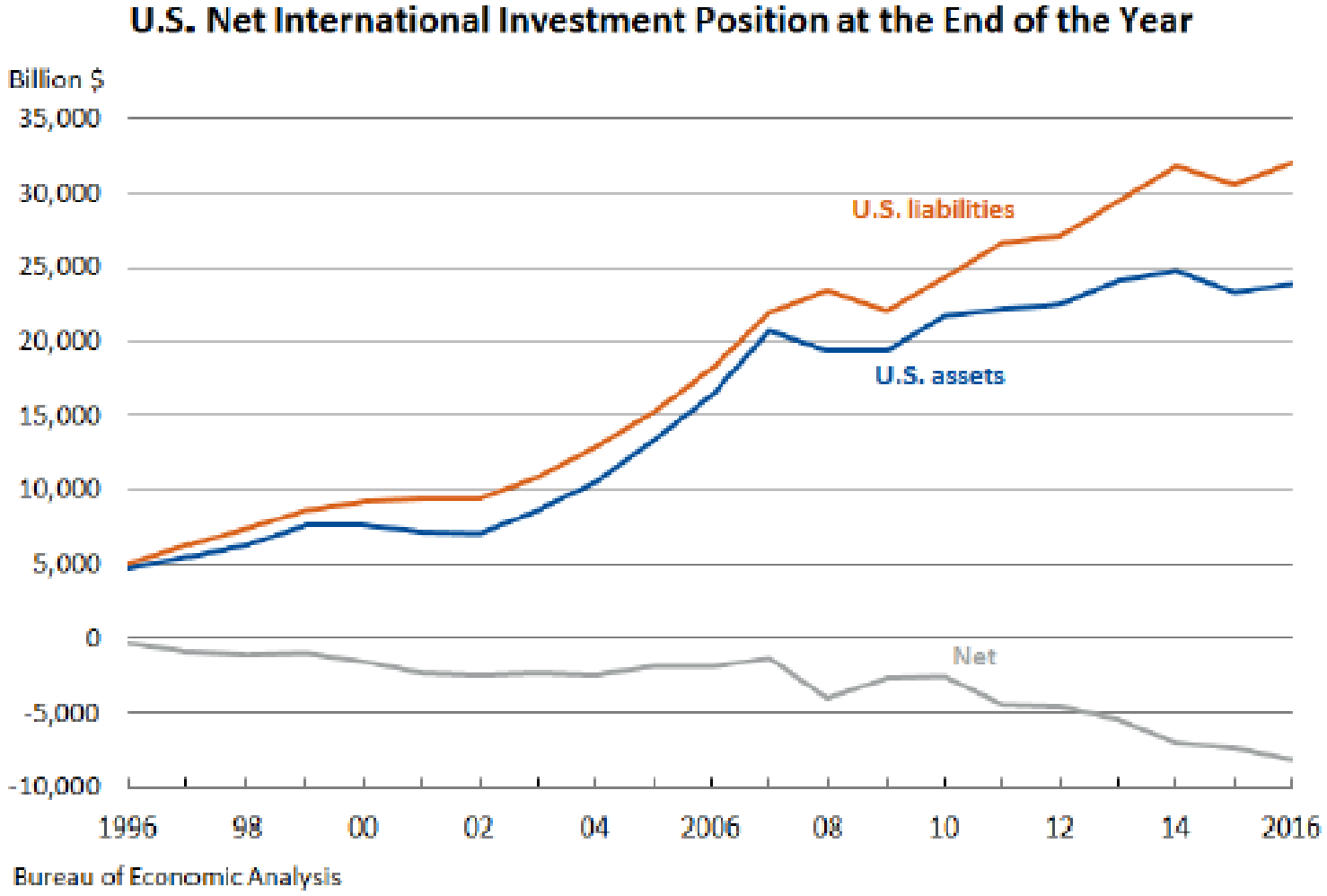
FIGURE 13-2

The U.S. Current Account and Net International Investment Position, 1976–2015

A string of current account deficits starting in the early 1980s reduced America's net foreign wealth until, by the early 21st century, the country had accumulated a substantial net foreign debt.

Source: U.S. Department of Commerce, Bureau of Economic Analysis.

CA and Net International Investment Position

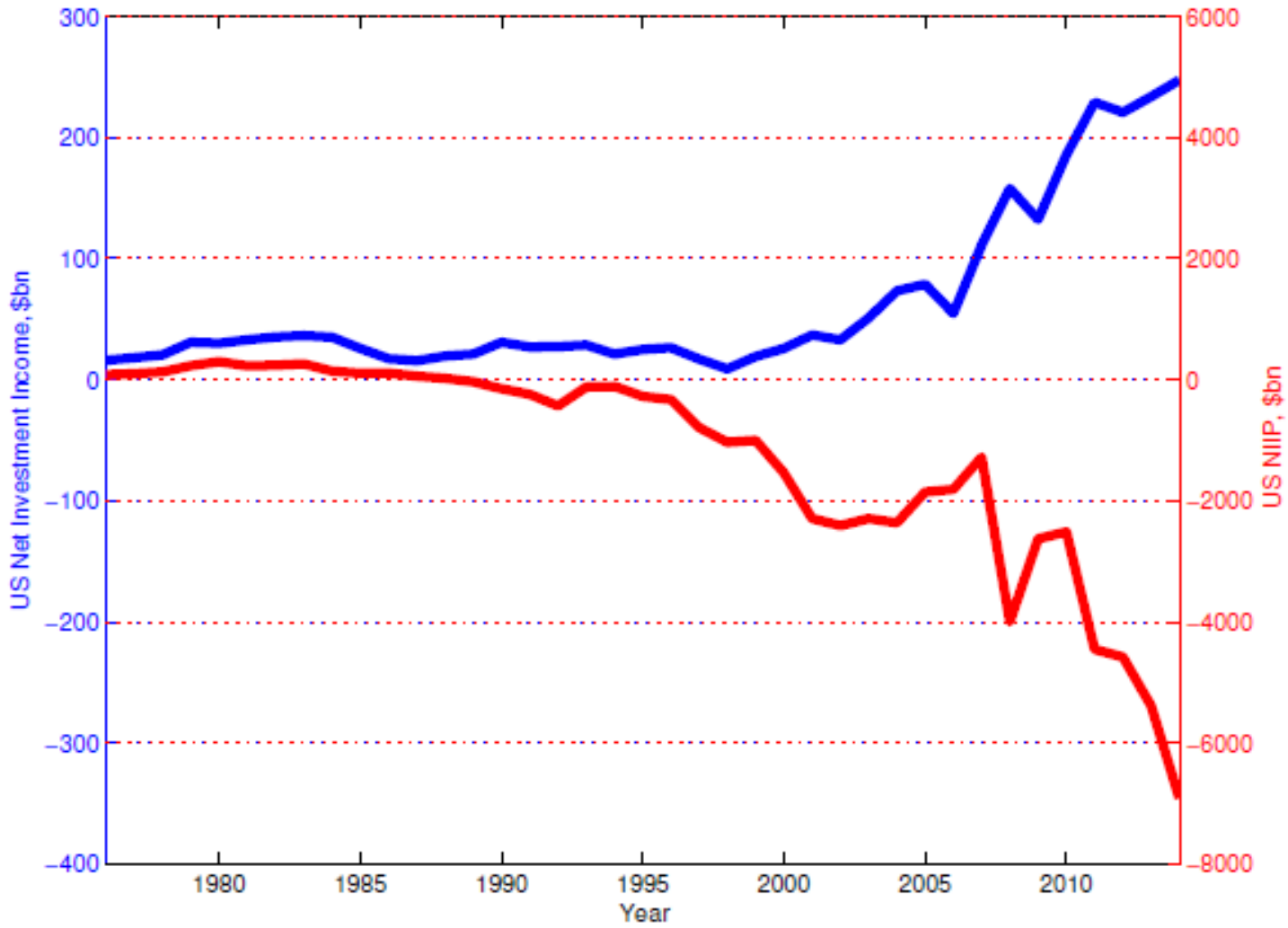


- Note the difference between *net* and *gross* **international investment position**

Question

If a country has a negative net international investment position, does it receive negative or positive return on this position?

Figure 1.7: Net Investment Income and the Net International Investment Position (United States 1976-2014)



Data Source: http://www.bea.gov

The Negative NIIP And Positive NII *Paradox*

- Net international investment position (NIIP)

$$NIIP = A - L$$

  – where A is country's foreign assets and L is country's foreign liabilities

- Net investment income (NII)

$$NII = r^A A - r^L L$$

  – where r^A is the interest on foreign assets, r^L is the interest on foreign liabilities

Gross International Investment Positions, U.S.

Right after WWII

1. NIIP/GDP=12%
2. Assets and liabilities are small
3. Assets>>Liabilities

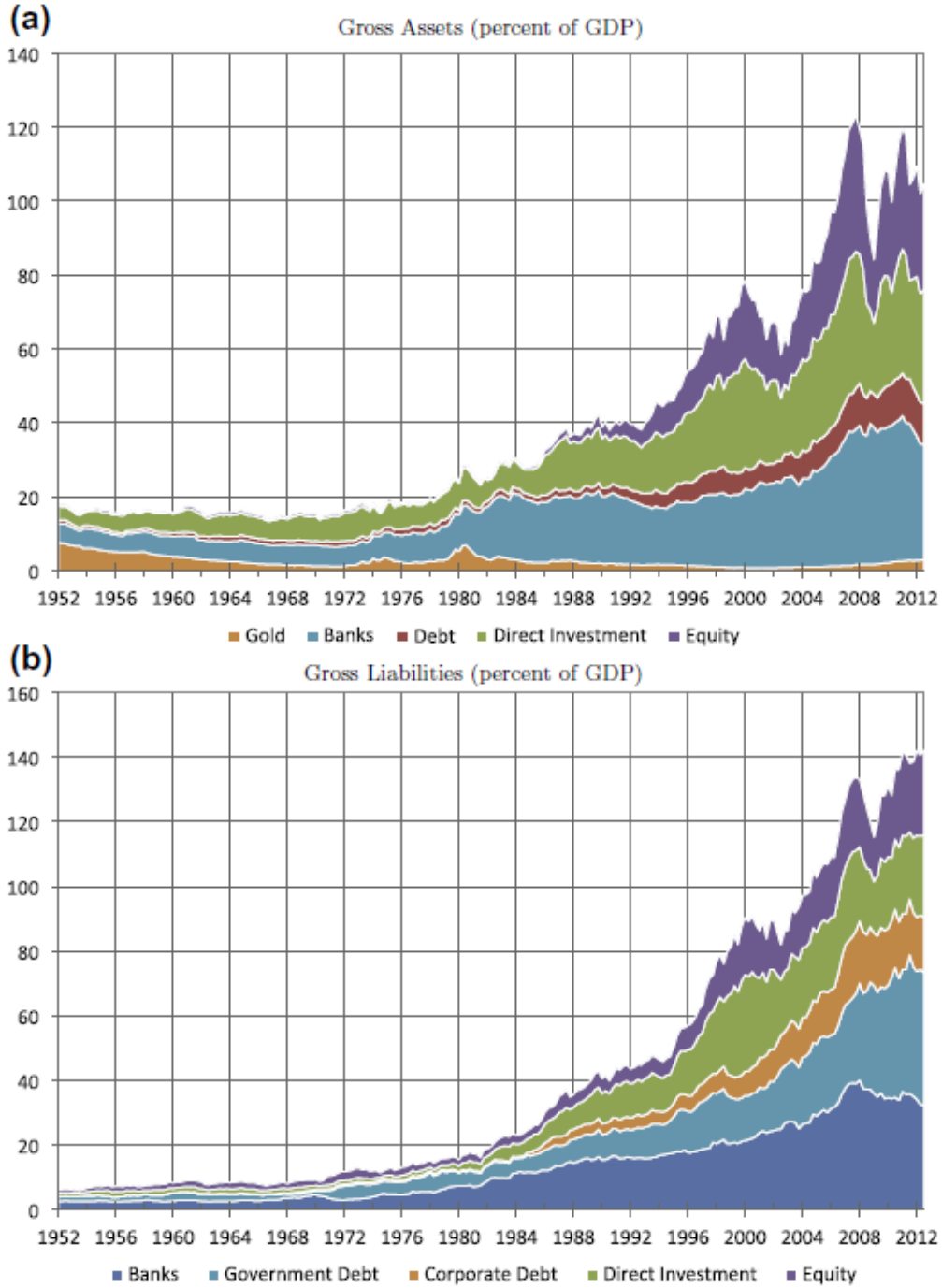


Figure 10.9 U.S. Gross Asset and Liabilities, by Asset Class, 1952–2012. *Source:* Gourinchas et al. (2010)

The 21st century

1. NIIP/GDP=-20% in 2010
2. Assets and liabilities are much bigger

Source: Gourinchas, Rey (2014) "External Adjustment, Global Imbalances, Valuation Effects" Section 5.1.3

US is like "a very leveraged investor"

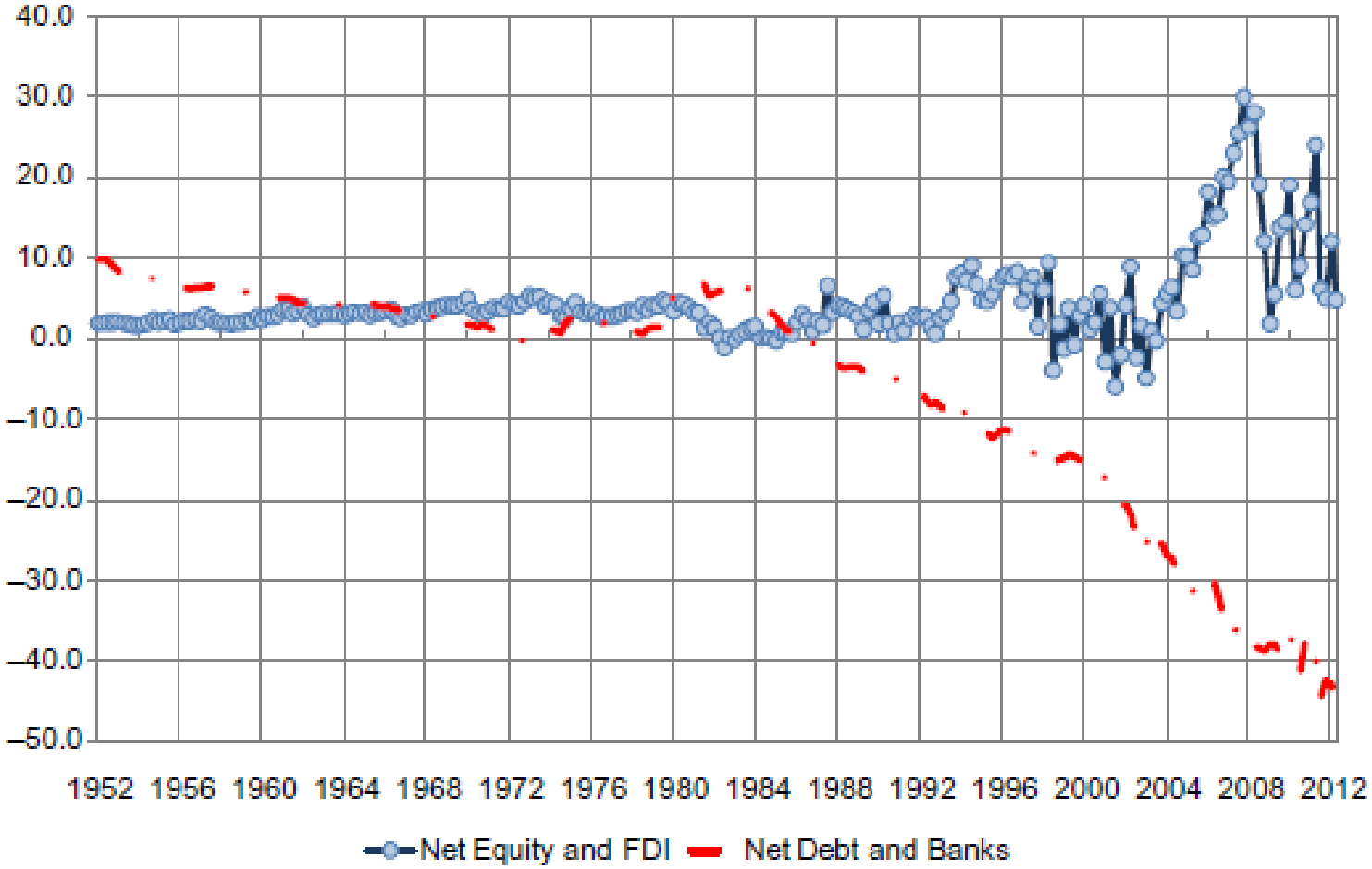


Figure 10.10 U.S. Net Portfolio Equity and Direct Investment (Percent of GDP) and Net Portfolio Debt and Other Assets (Percent GDP). *Source:* Gourinchas et al. (2010) Updated to 2012

- US is long on risky (*high* return) foreign assets and short on safe (*low* return) domestic assets
- The yield that the US received on its assets is higher than the yield that it paid on its liabilities by about 2% over 1952-2011. (This observation is often called ***exorbitant privilege***)